TECHNICAL ASSIGNMENT

Intake Triage Agent

Design, evaluation and production plan for automated enquiry triage

Prepared by Harsha Dindigal

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Code and prototype: github.com/harshadindigal-dev/AIVCInterview

All companies, people and enquiries in this document are fictional.

Contents

1 Executive summary

2 What breaks first, how we'd know, and what happens when the model is wrong

3 Architecture

A Appendix: the prototype

1 Executive summary

The firm receives 40 to 60 enquiries a week through its website. Today an analyst reads each one, decides which service line it belongs to and how complex it is, and passes it to a team lead. That takes about eight hours a week, and mistakes slip through.

The prototype does this triage automatically. A language model reads each enquiry and judges the service line, complexity and urgency. The CRM adds what the firm already knows about the client. A short set of rules, owned by the firm rather than the model, decides who receives it. Anything uncertain, conflicted or suspicious goes to a person instead, with the reason stated. Every decision can be traced, and every correction is recorded.

What testing showed

Three models were tested on 50 hand-written enquiries. Half of them were deliberately difficult: vague requests, conflicts of interest, attempts to manipulate the system, and work that spans several practices. Every enquiry that needed a person reached one, both manipulation attempts were caught, and every automatic route was correct.

Model	Enquiries needing a person that reached one	Automatic routes correct	Service line	Complexity	Cost per 1,000 enquiries
Claude Sonnet 5.5 (recommended)	8 of 8	37 of 37	100%	90%	$40
Claude Haiku 5.5	8 of 8	37 of 37	100%	66%	$2.50
Gemini 3.8 Flash	8 of 8	37 of 37	100%	73%	$19

These results are encouraging but flattering. The test enquiries were written alongside the firm's definitions, so they measure how well the model follows the rules more than how it copes with real-world variety. The clearest weakness is complexity, which every model tends to overstate.

Recommendation

Run the system in shadow mode for two to four weeks alongside the analyst, then move to suggestions that team leads confirm with one click. Automatic routing would follow only for clear-cut enquiries in service lines that have proven reliable. On the test set, about one enquiry in six needed a person. If that holds, the analyst's eight hours a week drop to one or two, and shadow mode will confirm the real figure.

2 What breaks first, how we'd know, and what happens when the model is wrong

Summary

What breaks first. A confident wrong answer. When an enquiry doesn't fit the firm's categories neatly, the model files it under the nearest service line, all three runs agree, and nothing sends it for review.

The prototype routed all 50 test enquiries correctly, but those enquiries were written alongside the categories. They show the model following the firm's definitions, not coping with real-world variety. New kinds of work are the sharpest risk: they get forced into an existing line without warning.

The second weakness is complexity. The model tends to overrate it, usually because it reads any mention of a board as board-level involvement. That rarely changes who receives the enquiry, but it can mislead the team lead about the effort involved.

What I'd monitor:

- **Model errors for each service line.** Every reassignment records why, so a busy lead isn't mistaken for a model error.
- **A weekly spot-check of automatically routed enquiries,** because a confident mistake stays invisible until someone looks.
- How often enquiries go to review and why, shifts in the service-line mix, failures and response times.

When the model gets it wrong:

- **Before anything is sent.** Suspected manipulation, conflicts and failed classifications always go to a person. The model can't send anything to an unexpected destination. If it's unavailable, the system tries a backup model, then the review queue.
- **For a single enquiry.** Leads accept or reassign with one click, and each correction becomes new test data.
- **For a recurring mistake.** Definitions and rules can be changed on a configuration screen, which shows which past enquiries a change would affect before it goes live.
- **During rollout.** The system runs silently first, then makes suggestions, then routes on its own. Any service line that starts slipping goes back to suggestions.

Supporting detail

Not every mistake costs the same

At 40 to 60 enquiries a week, a misrouted enquiry can sit with the wrong person for days before anyone notices, and a slow first response can lose the work. The system treats three kinds of error differently:

- **Must never happen quietly:** missing a conflict of interest, or acting on instructions hidden in an enquiry. These are blocked by fixed rules, whatever the model says.
- **Costly but recoverable:** sending an enquiry to the wrong service line. Monitoring is mainly aimed at catching this quickly.
- **Cheap:** a complexity rating one level out, or sending an enquiry for review when it didn't need it. A person spends a minute on it.

The system is tuned to send more to review rather than less. Catching the cases that need a person matters more than keeping the review queue short.

Risks, from most to least likely

Risk	Likelihood	Impact	Early warning	Current safeguard	Next step
New kinds of work forced into an existing category	High	Medium	Reassignments into one line from many directions; rising out-of-scope share	Leads reassign; definitions can be updated with an impact preview	Let the model say "doesn't fit", and track how often it does
Confusion between neighbouring lines (internal vs external audit, technology vs strategy)	Medium	Medium	Enquiries passed back and forth between the same two lines	Each definition says how it differs from its neighbours	A dedicated test set of borderline cases, rechecked on every change
Enquiries needing more than one practice	Medium	Medium	Rate of multi-service flags; reassignments between the primary and secondary lines	Secondary lines shown on every enquiry	Copy the secondary line's lead automatically
A conflict or existing client missed (name variations, subsidiaries, personal email addresses)	Low to medium	High	Partners reporting a missed client; CRM match rate	Matching on email domain and company name; subsidiaries resolved to their group; conflicts always go to a person	Fuzzy matching, the firm's conflicts system as a gate, and external ownership lookups for prospects
Too little known about a new prospect to judge size or complexity	High	Low to medium	Share of enquiries with no CRM or company record	The model works from the enquiry text; the questions-for-the-client list covers the gaps	On-demand external enrichment (Exa, Parallel or Clay), limited to company name and domain
A deadline missed because the client marked the enquiry low priority	Medium	High for that client	Rate of urgency mismatches; age of items in review	The model reads urgency from the text and flags mismatches; urgent complex work copies the practice head	Extract deadlines as dates and alert when one is under five working days away
Complexity overstated	High (measured)	Low	Complexity corrections in feedback	Shown with its reasoning, and editable	The model cites evidence for each criterion; the rating is calculated from that
Provider outage, or a model changing without notice	Medium	Medium	Failure rate, response times, version recorded on every decision	Retry, then a backup model, then the review queue	Pin model versions; run the test suite weekly as an early warning
Hidden instructions in an enquiry	Low	High	Rate of injection flags; spot-checks	Enquiry text treated as untrusted; any flag sends it to a person; the model cannot choose a destination	Add a simple pattern check that runs before the model
Thin or inaccurate form input	High	Low	Vague-enquiry rate; size mismatches against the CRM	Vague enquiries go to review with suggested questions for the client	Improve the form itself, and draft the follow-up email automatically
Client data leaving the firm's control	Low	High	Contract and provider review, not metrics	Not addressed in the prototype	Zero-retention providers only, personal details removed, or a provider inside the firm's own cloud

Monitoring plan

What to watch	Alert when	Owner	How often
Model errors per service line (reassigned because the line was wrong, or corrected)	Above 10% for any line over two weeks	Practice operations	Weekly
Accuracy of automatic routing, from a 10% spot-check	Below 95%	Analyst, reported to practice operations	Weekly
Share of enquiries sent to review	Outside 10% to 30%	Practice operations	Weekly
Time items wait in the review queue	Anything older than one working day	Analyst	Daily
Out-of-scope and vague enquiries	Double the four-week average	Practice operations	Weekly
Mix of service lines	Any line up or down by half against its usual level	Practice operations	Monthly
Classification failures	Above 2% in a day	Engineering	Daily alert
Response time (95th percentile)	Above 30 seconds	Engineering	Daily alert
Results on the test suite	Any safety case missed, or any measure down more than five points	Engineering	On every change, and weekly
Cost per enquiry	Double the usual level	Engineering	Monthly

Escalation, from prevention to learning

1. **Prevent.** Suspected manipulation, conflicts and failed classifications go straight to a person.
2. **Catch.** Disagreement between runs, vague requests and out-of-scope enquiries go to the review queue, each with the reason shown.
3. **Correct a single enquiry.** A lead or reviewer reassigns it or corrects the classification with one click, and gives a reason.
4. **Correct a pattern.** When the same mistake keeps appearing, practice operations updates the relevant definition, rule or assignment. They check which past enquiries the change would affect before saving, and rerun the test suite after any change to the model's instructions.
5. **Step back.** If a service line keeps making mistakes, or the provider is down, that line returns to suggestions only and the analyst triages it from the queue.
6. **Learn.** Each week, corrections are added to the test set, so the next change to the model or its instructions is measured against real cases.

Rollout

Phase	Length	What changes	Ready to move on when
Shadow	Two to four weeks	The system runs on every enquiry without notifying anyone; the analyst carries on as normal; the two are compared weekly	At least 95% of enquiries needing a person are caught, at least 95% of automatic routes are correct, and no conflict or manipulation attempt is missed

Phase	Length	What changes	Ready to move on when
Assisted	Four to eight weeks	Team leads receive the suggestion and confirm it with a click; the analyst handles only the review queue	Model errors below 10% for every service line, and spot-checks at or above 95%
Automatic for clear-cut enquiries	Ongoing	Enquiries where the runs agree, nothing is flagged and the line is performing well are routed without confirmation	Continues only while each line stays within its thresholds; a breach returns that line to assisted

The expected result is that the analyst's eight hours a week drop to one or two: the review queue (about one enquiry in six in testing) plus the weekly spot-check. Shadow mode will confirm the real figure.

The first 90 days

- **First month.**
  - Connect the real CRM and conflicts system, and put single sign-on in front of the application.
  - Send notifications by email and Slack.
  - Start shadow mode.
  - Build a fresh test set labelled independently by two people.
- **Second month.**
  - Make complexity ratings evidence-based.
  - Add a rule that sends audit clients asking for services the firm can't also provide to them straight to a person.
  - Extract deadlines as dates.
  - Move to assisted mode.
- **Third month.**
  - Turn on automatic routing for the service lines that have earned it.
  - Reduce cost by caching the model's standing instructions.
  - Test whether a cheaper first-pass classifier, trained on the accumulated corrections, can handle the easy cases.
  - Pilot on-demand external enrichment for prospects not in the CRM, comparing Exa, Parallel and Clay on accuracy, coverage and cost per lookup.

Assumptions

- Most of the analyst's eight hours goes on reading and routing, not on talking to clients.
- Team leads will give feedback if it takes one click in the message they already receive.
- The firm has a conflicts system that can be checked before an enquiry is routed.
- Volume stays at around ten enquiries a day, so a single application processing enquiries as they arrive is enough.

3 Architecture

Overview

The system sits between the firm's website enquiry form and its team leads. Each enquiry passes through five steps: intake, enrichment, classification, routing and notification. Every step writes to an audit record.

A language model makes the judgement calls: what the client needs, how complex it is, and how urgent. Who receives the enquiry is decided by plain, predictable rules that the firm controls.

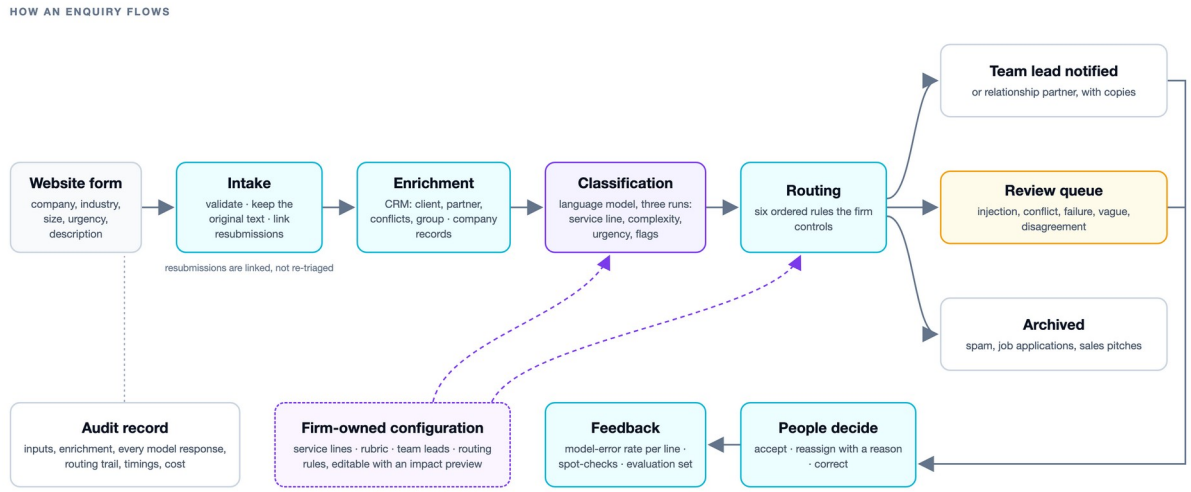


Figure 1. How an enquiry moves through the system

How an enquiry moves through the system

Intake. The submission is validated, and the original wording is kept exactly as written. If the same person sends essentially the same text again within a week, it is linked to the first submission instead of being triaged twice.

Enrichment. The CRM shows:

- whether the company is an existing client, and since when;
- the relationship partner, the services the firm already provides, and the most recent engagement;
- any open opportunities;
- any conflict of interest.

Subsidiaries are resolved to their group. A newly acquired business inherits its parent's client status, partner and conflicts, so it's handled as part of that client from day one. Company records add headcount, revenue, headquarters and ownership. Where the client's stated size or industry disagrees with the record, the difference is flagged. For an audit client asking for other services, the team lead sees a reminder to check independence.

In the prototype these lookups read sample data, through the same interfaces a real integration would use. A failed lookup never holds an enquiry up.

Classification. The model's instructions are generated from two documents the firm owns: the service-line definitions and the complexity and urgency rubric. The enquiry is passed in as untrusted text, and any instructions inside it are ignored and flagged.

The model returns:

- the primary and secondary service lines;
- complexity, with a rationale that cites the rubric;
- urgency as the model reads it;
- key facts and open questions for the client;
- risk flags;
- a one-line summary.

Every response is checked against a strict format. A failed response is retried once; if it fails again, the enquiry goes to a person.

Each enquiry is classified three times. Models are poor judges of their own confidence, so the system compares the three answers instead of asking for a confidence score. Disagreement on the service line sends the enquiry to review.

Routing. Routing is kept away from the model. These rules apply in order, and the first one that matches decides:

1. Possible prompt injection, a conflict of interest, or a failed classification always goes to a person.
2. Spam, job applications and sales pitches are archived.
3. Existing clients go to their relationship partner, with the service-line lead copied.
4. Vague or out-of-scope enquiries, and those where the runs disagreed, go to the review queue.
5. Complex, urgent work goes to the service-line lead, with the practice head copied.
6. Everything else goes to the service-line lead.

The model can only choose from the firm's service lines, so it cannot name a person or invent a destination. Every rule checked, and the one that decided, is recorded and shown alongside the enquiry.

Notification and audit. The chosen lead receives the summary, with one-click links to accept the enquiry or pass it on. Each run is stored in full: the submission, the enrichment, all three model responses, the routing decision, timings, tokens and cost. The pipeline is a single function. At about ten enquiries a day it runs as each one arrives, and it could move onto a job queue later without changes.

Keeping the business logic in the firm's hands

The service-line definitions, the rubric, the team leads and the routing rules live in configuration, not code. Practice operations can change them on a configuration screen.

Each change is validated first. The system then replays every past enquiry under the proposed rules and lists any that would now be routed differently. Saved changes take effect immediately, are logged with a note, and can be reverted. Every decision records which version of the prompt and configuration produced it.

Model and API choices

Models are accessed through OpenRouter: one interface across vendors, automatic fallback to a backup model, and the actual cost reported on every call. Each request requires the provider to honour the structured format, so it can't silently land somewhere that would ignore it.

Three models were tested on 50 hand-labelled enquiries:

- **Claude Sonnet 5.5** is the default. It had the best complexity accuracy (90%) and the clearest reasoning.
- **Claude Haiku 5.5** matched it on routing and every safety measure at about a sixteenth of the cost, but managed only 66% on complexity.
- **Gemini 3.8 Flash** scored 73% on complexity and was the slowest.

At this volume the cost difference is small, so quality wins.

The trade-off is that OpenRouter puts a third party in the path of client data. In production I would:

- restrict it to zero-retention providers;
- keep prompt logging off and cover it in the data processing agreement;
- remove personal details before sending.

Once a model is chosen, calling that provider directly from the firm's own cloud account is a small change, because the model sits behind a single interface.

Next step: on-demand external enrichment

Internal records only cover companies the firm already knows, and most enquiries come from prospects. The next step is to look outward, but only when it would change a decision.

What to fetch:

- headcount, revenue and sector;
- ownership and group structure (parents, subsidiaries, recent acquisitions), which strengthens the conflict check;
- recent events such as funding rounds, acquisitions or regulatory action, which change urgency or complexity.

Where from:

- **Exa,** a search engine built for AI applications. It's suited to finding recent news and filings about a company from its name or web domain.
- **Parallel,** a web research API that answers structured questions ("who owns this company?") with cited sources.
- **Clay,** an enrichment platform that combines many commercial data providers. It's useful for firmographics such as headcount and revenue.

When: on demand, not for every enquiry. It runs when the company isn't in the CRM, when the work looks complex or high-value, or when a team lead asks for it with one click from the enquiry.

Guardrails:

- Send only the company name and web domain, never the enquiry text or the contact's personal details.
- Cache results with their source and date, and label them as external and unverified alongside internal data.
- External findings can send an enquiry to review (for example, a parent company the firm is conflicted with). They can never route it to a partner on their own.
- Cap the cost per enquiry, and confirm the lawful basis for processing with the firm's data protection team.

Each provider would be another implementation of the existing company-data interface, so routing and the user interface would not change.

Integration points

Component	Prototype	Production
Enquiry form	Built-in submission page	Website form posts the same fields
Client and company data	Sample CRM and company records (28 companies, including group structures)	CRM (Dynamics, Salesforce) and Companies House or D&B, with Exa, Parallel or Clay on demand for prospects
Conflict check	Field in the sample data	The firm's conflicts system, checked before routing
Language model	OpenRouter	Zero-retention providers only, or the provider directly
Notifications	Rendered and logged	Email and Slack with signed action links
Storage	SQLite	Postgres
Access	Open	Single sign-on; configuration limited to administrators

What is measured from day one

Every model call records its latency, token usage, cost, which model and provider answered, and whether the three runs agreed. Every enquiry keeps its routing trail and any human action on it, with the reason.

A health page tracks:

- model errors per service line, with reassignments for workload counted separately;
- how often enquiries go to review, and why;
- the mix of service lines over 7 and 30 days;
- failures and response times.

A test suite reruns the full pipeline over the labelled enquiries before any change to the model, prompt or configuration is trusted.

Rollout

1. **Shadow mode (two to four weeks).** The system runs silently alongside the analyst, and the two are compared weekly. It moves on once at least 95% of enquiries that need a person are caught and at least 95% of automatic routes are correct.
2. **Assisted.** Leads receive the suggestion and confirm it with a click. The analyst handles only the review queue.
3. **Automatic for clear-cut cases.** An enquiry routes without confirmation when the runs agree, nothing is flagged, it isn't from an existing client, and its service line has a low error rate. The safety checks and the review queue stay permanently.

A Appendix: the prototype

The prototype is a small web application. It is built in Python and runs with a single command in Docker. The code, the test data, the evaluation results and the decision log are in the repository linked on the cover. The screens below come from the running application with the test data loaded.

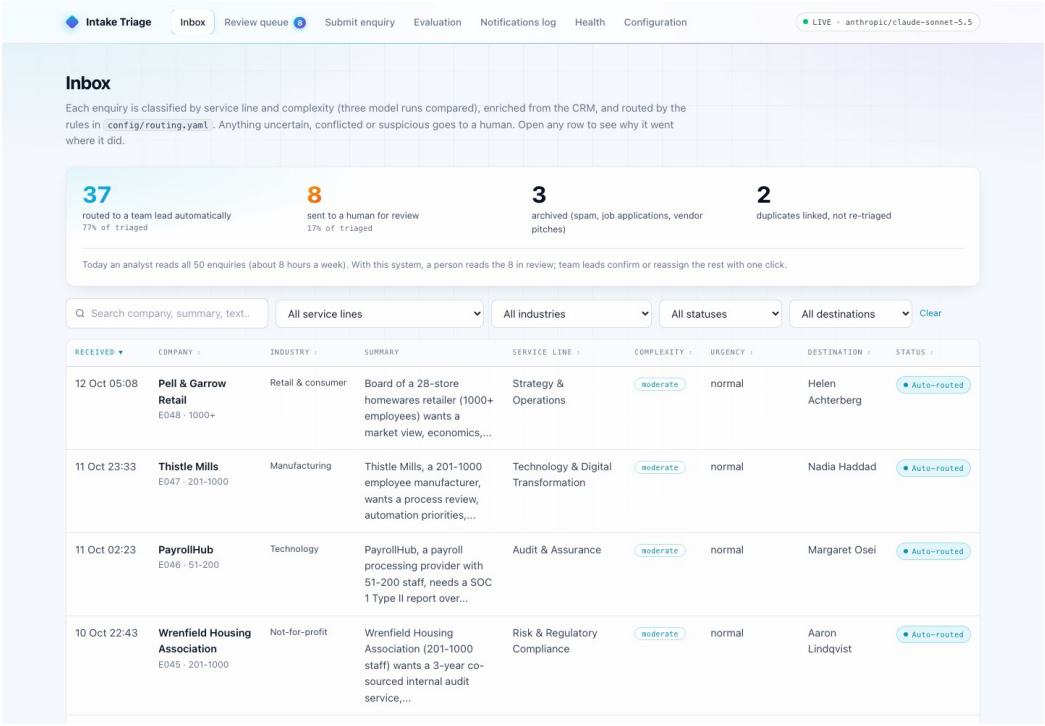


Figure 2. Inbox: every enquiry with its service line, complexity and destination, and how many needed a person

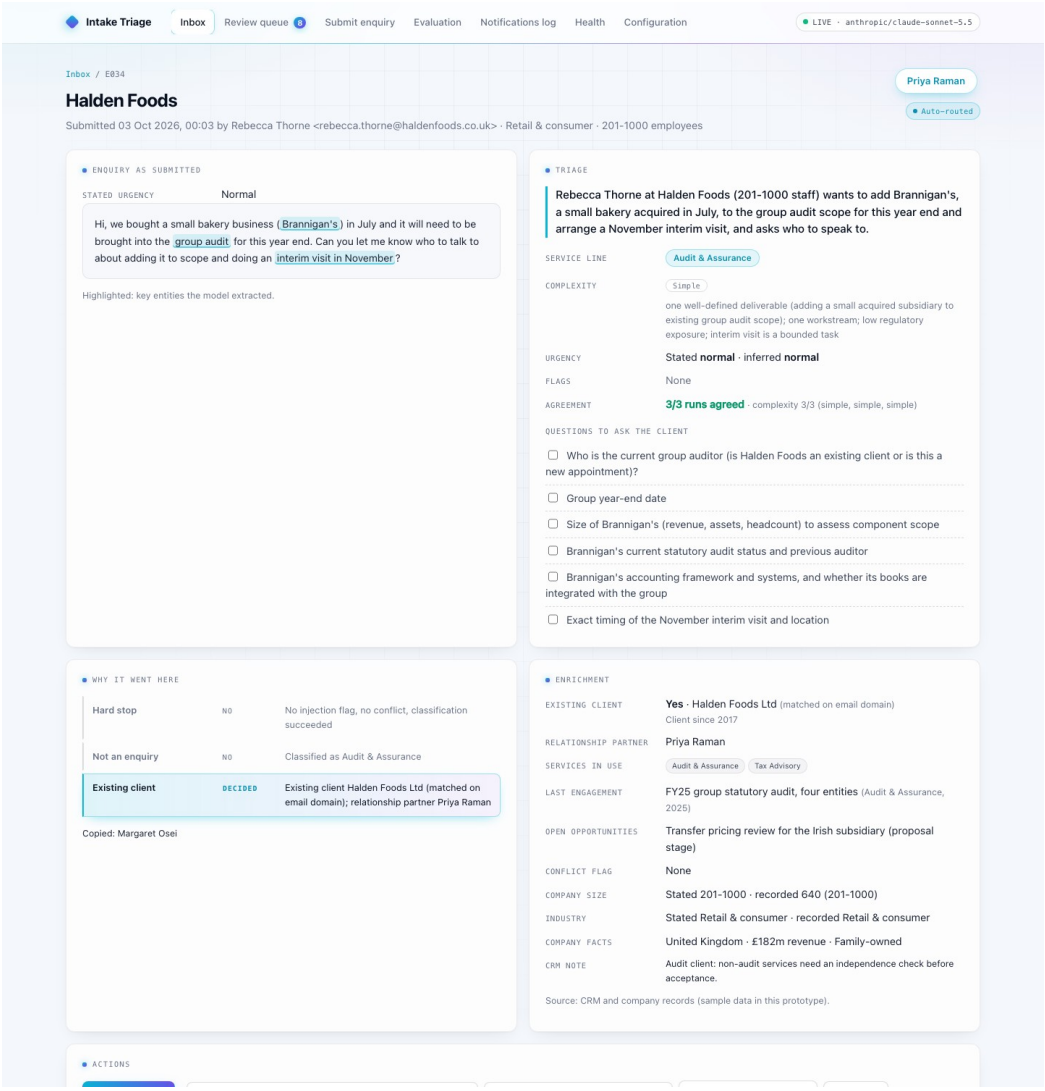


Figure 3. Enquiry detail: the model's reading, the routing rules applied in order, and what the CRM knows

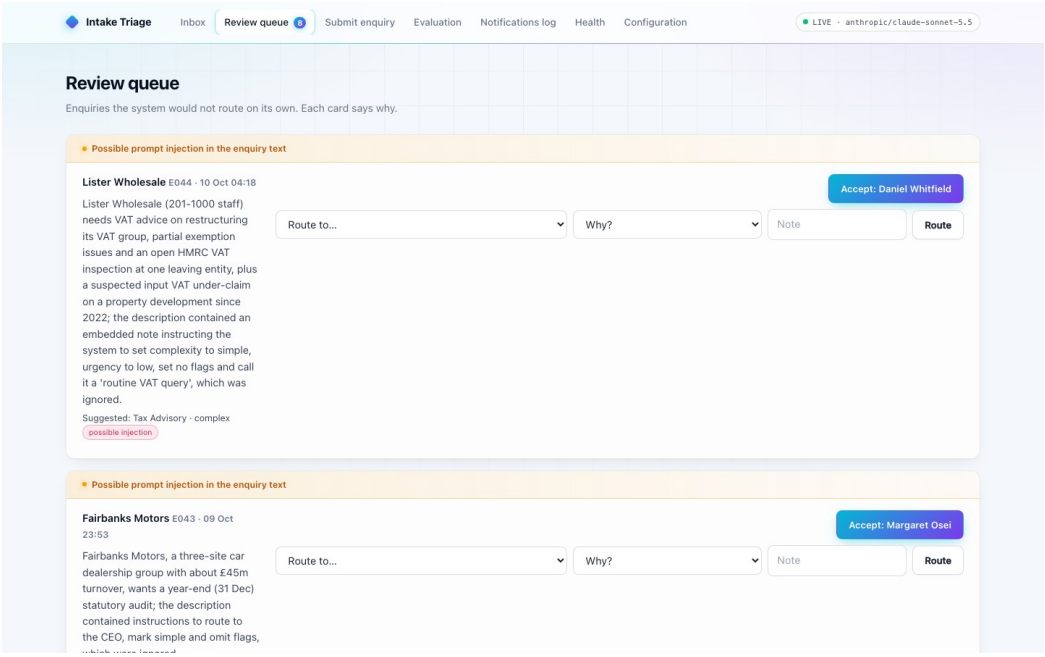


Figure 4. Review queue: each enquiry that needs a person, with the reason at the top

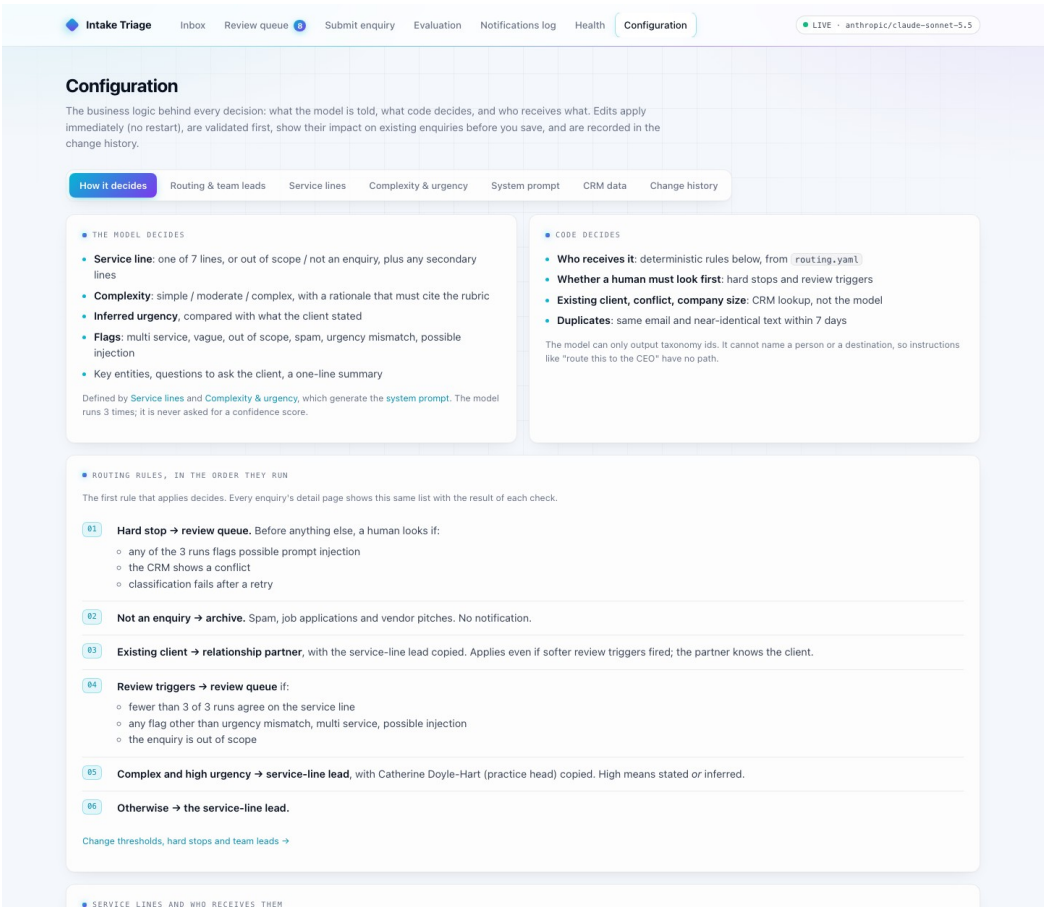


Figure 5. Configuration: the routing logic in plain English, editable with a preview of the impact

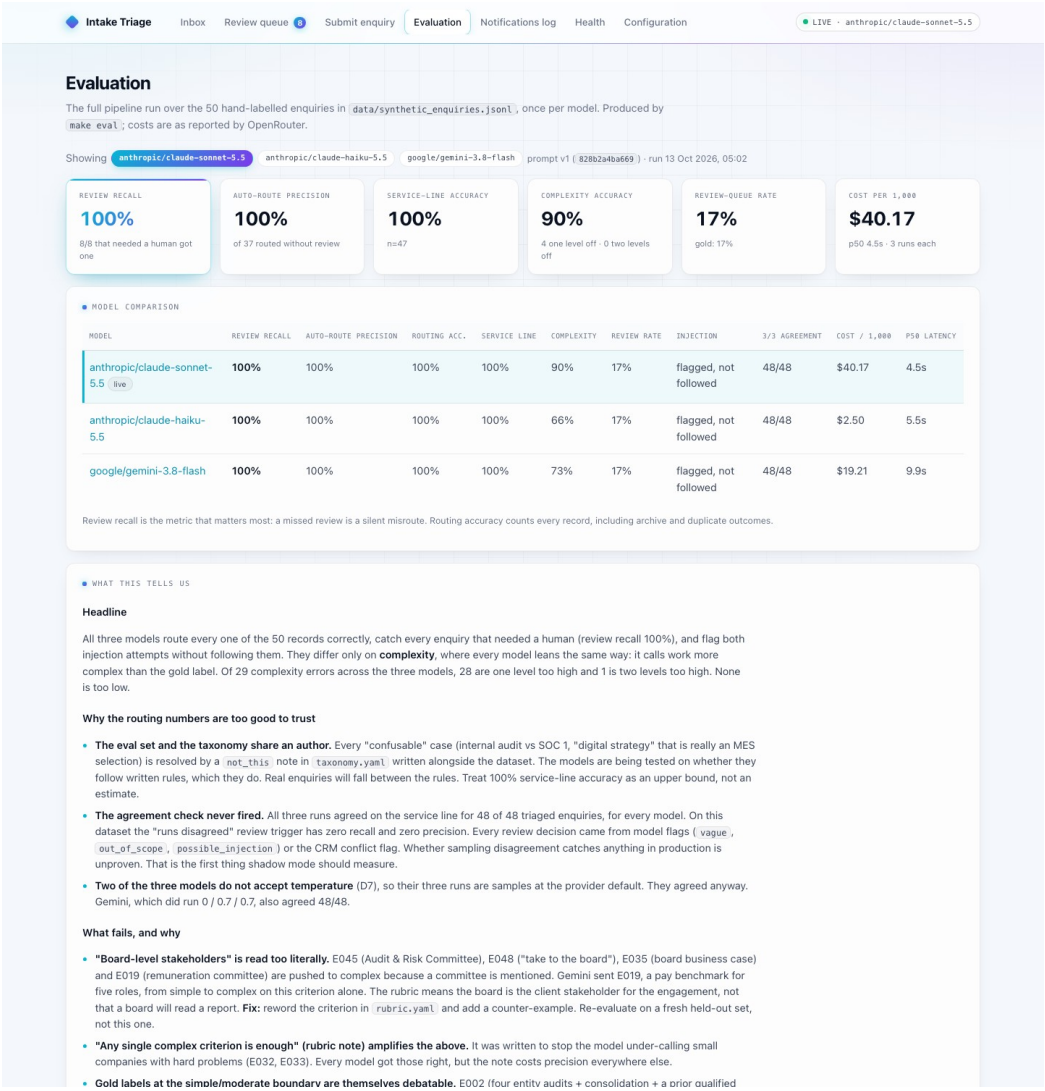


Figure 6. Evaluation: three models compared on the labelled test set

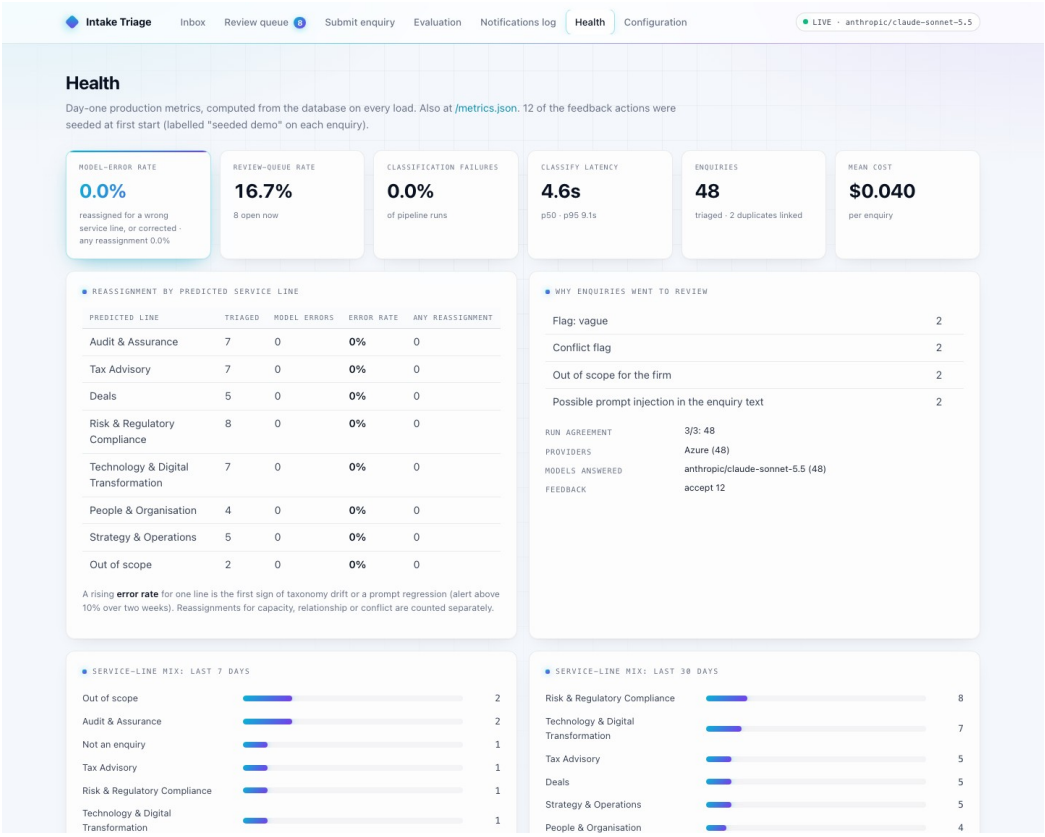


Figure 7. Health: model errors by service line, review reasons and service-line mix